Formatting Instructions For NeurIPS 2026

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Abstract

The abstract paragraph should be indented ½ inch (3 picas) on both the left- and right-hand margins. Use 10 point type, with a vertical spacing (leading) of 11 points. The word **Abstract** must be centered, bold, and in point size 12. Two line spaces precede the abstract. The abstract must be limited to one paragraph.

1 Submission of papers to NeurIPS 2026

Please read the instructions below carefully and follow them faithfully.

1.1 Style

Papers to be submitted to NeurIPS 2026 must be prepared according to the instructions presented here. Papers may only be up to **nine** pages long, including figures. **Papers that exceed the page limit will not be reviewed (or in any other way considered) for presentation at the conference.** Additional pages *containing acknowledgments, references, checklist, and optional technical appendices* do not count as content pages.

The margins in 2026 are the same as those in previous years.

Authors are required to use the NeurIPS LaTeX style files obtainable at the NeurIPS website as indicated below. Please make sure you use the current files and not previous versions. Tweaking the style files may be grounds for desk rejection.

1.2 Retrieval of style files

The style files for NeurIPS and other conference information are available on the website at

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The only supported style file for NeurIPS 2026 is `neurips_2026.sty`, rewritten for LaTeX 2_ε. **Previous style files for LaTeX 2.09, Microsoft Word, and RTF are no longer supported.**

The LaTeX style file contains three optional arguments:

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The formatting instructions contained in these style files are summarized in Sections **??**, **??**, and **??** below.

2 Methods

We systematically evaluated four prompting and retrieval strategies to understand their impact on medical question answering performance:

(1) Baseline Prompting

We used standard direct prompting without any additional reasoning scaffolding or retrieval augmentation. Each question was passed to the model as-is, and the model generated an answer in a single step. This serves as a control condition to benchmark improvements from more advanced methods.

(2) Chain-of-Thought (CoT) Prompting

To encourage step-by-step reasoning, we augmented prompts with instructions that explicitly guide the model to produce intermediate reasoning steps before arriving at a final answer. This method is particularly relevant for medical QA tasks that require multi-hop reasoning or interpretation of clinical evidence.

(3) Meta Prompting

Meta prompting introduces higher-level instructions that guide the model on *how* to approach the task (e.g., "analyze carefully," "consider multiple possibilities," or "verify before answering"). This approach aims to improve robustness and reduce hallucination by shaping the model's reasoning strategy rather than just its output format.

(4) MedRAG-based Retrieval-Augmented Generation (RAG)

We implemented a domain-specific RAG pipeline (MedRAG), where relevant biomedical documents are retrieved and incorporated into the prompt as contextual evidence. The model then generates answers grounded in the retrieved information. This method is designed to improve factual accuracy and reduce reliance on parametric knowledge alone.

2.1 MedRAG Modifications

To enhance the standard RAG pipeline for medical QA, we introduced three key modifications:

(1) Adaptive-*k* Retrieval Based on Query Complexity

Instead of retrieving a fixed number of documents (k), we dynamically adjust k based on estimated query complexity.

- Simpler factual questions use smaller k to reduce noise.
- Complex or multi-hop questions retrieve more documents to increase coverage.
  Query complexity is estimated using heuristic features such as question length, presence of multiple entities, and syntactic structure.

(2) Citation-Aware Generation with Structured Prompting

We enforce a structured prompt format where retrieved snippets are clearly labeled and the model is instructed to cite evidence when generating answers. This improves interpretability and encourages grounding:

- Each retrieved passage is tagged (e.g., [Doc1], [Doc2])

- The model is prompted to explicitly reference supporting documents in its reasoning or final answer
  This mechanism helps reduce hallucination and enables answer verification.

(3) Safer Snippet Extraction in the Retrieval Pipeline

To improve the quality and reliability of retrieved context, we refine snippet extraction by:

- Filtering out incomplete or low-confidence passages
- Removing noisy or irrelevant text segments
- Ensuring biomedical relevance using domain-specific heuristics
  This step reduces the likelihood of misleading context being passed to the model.

2.2 3. Experimental Setup

We used **GPT-4o** as the base model for all experiments to ensure consistency across methods.

Datasets

We evaluated performance on four widely used medical QA benchmarks, covering different question formats:

- **MedQA-US**: Multiple-choice questions (clinical knowledge assessment)
- **PubMedQA**: Yes/No/Maybe questions based on biomedical abstracts
- **BioASQ**: Factoid questions requiring concise answers
- **MediQA**: Open-ended responses evaluated via answer similarity

For each dataset, we sampled approximately **250 questions**, resulting in a total of $\tilde{1}$,000 evaluation examples.

Evaluation Protocol

Each method (Baseline, CoT, Meta Prompting, MedRAG, and MedRAG variants) was applied consistently across all datasets. To ensure fairness:

- Prompts were standardized except for method-specific modifications
- Retrieval (for RAG methods) used the same corpus and embedding pipeline
- All outputs were generated with identical decoding settings

2.3 Evaluation Metrics

We evaluated model performance along three dimensions:

(1) Objective Accuracy

For datasets with discrete ground-truth answers (e.g., MedQA, PubMedQA), we computed exact match accuracy.

(2) Subjective Quality (BERTScore F1)

For open-ended responses (e.g., MediQA), we used BERTScore F1 to measure semantic similarity between generated answers and reference answers. This metric captures meaning-level alignment beyond exact wording.

(3) Hallucination Assessment

We performed answer-level hallucination evaluation, where outputs were manually or programmatically judged based on whether:

- Claims are supported by retrieved evidence (for RAG methods)
- Content is factually consistent with known medical knowledge
  This provides insight into model reliability and safety, which is critical in medical applications.

This experimental framework allows us to:

- Compare prompting strategies vs. retrieval-based approaches
- Evaluate the effectiveness of MedRAG enhancements
- Analyze trade-offs between accuracy, reasoning quality, and hallucination

By combining controlled experiments with multiple evaluation metrics, we aim to provide a comprehensive assessment of LLM performance in medical QA settings.

3 Results

3.1 Accuracy

Table 1 reports exact-match accuracy on the two objective benchmarks. On MedQA-US, Chain-of-Thought (CoT) prompting achieves the highest accuracy (0.8783), outperforming both the Baseline (0.8174) and all MedRAG variants, which match the Baseline exactly. Meta prompting degrades accuracy to 0.7609, the lowest of any condition. On PubMedQA the picture inverts: Baseline, CoT, and Meta all cluster near chance (0.22–0.25), while both MedRAG variants reach 0.6400, a jump of roughly 0.39 over Baseline. The gap highlights a fundamental difference between the two tasks: MedQA-US rewards parametric reasoning over clinical options, whereas PubMedQA requires evidence-backed binary decisions that retrieval grounding directly supports.

Table 2 reports BERTScore F1 on the two subjective benchmarks. On BioASQ, MedRAG adaptive-k achieves the highest score (0.9070), with fixed-k close behind (0.9049), both substantially above Baseline (0.8480). CoT underperforms Baseline here (0.7846), suggesting that verbose step-by-step reasoning conflicts with the terse factoid format BioASQ expects. On MediQA all five conditions lie within a narrow band (0.8203–0.8373), with Baseline performing best. The near-uniform BERTScore on MediQA suggests that the candidate-set answer format reduces discriminative power, and differences between conditions are unlikely to be meaningful.

3.2 Hallucination

Table 3 reports answer-level hallucination rates across all four datasets. Figure 3.2 visualizes these rates, and Figure 3.2 shows raw hallucinated counts; per-dataset verdict breakdowns are presented in Figures 3.2–3.2.

PubMedQA. Both MedRAG variants reduce hallucination substantially relative to Baseline (0.4200): adaptive-k reaches 0.1760 and fixed-k reaches 0.1880, with CoT at 0.2040. The PubMedQA verdict breakdown (Figure 3.2) makes the mechanism visible: Baseline and Meta carry a large hallucinated segment ($\approx$105 out of 250), whereas MedRAG nearly halves it to $\approx$44–51, and the proportion of *supported* responses rises sharply.

BioASQ. CoT achieves the lowest hallucination rate by a large margin (0.0147 vs. Baseline 0.0882), a six-fold reduction. MedRAG fixed-k provides a moderate improvement (0.0588), but adaptive-k (0.0809) barely outperforms Baseline, and Meta matches Baseline exactly. The BioASQ verdict breakdown (Figure 3.2) shows CoT produces almost no hallucinated sentences, but at the cost of shifting many responses to *unverifiable*—an important nuance discussed in Section 4.

MedQA-US. CoT again achieves the best hallucination rate (0.0960). All MedRAG variants perform at or above Baseline (0.1720–0.1800 vs. 0.1680), meaning retrieval offers no hallucination benefit on this multiple-choice task. Given that MedQA-US does not require the model to generate factual claims (only to select an option and briefly justify it), this result is not surprising.

MediQA. CoT achieves the best hallucination rate (0.0080). However, both MedRAG variants (0.0480, 0.0600) are substantially *worse* than Baseline (0.0240), and even worse than Meta (0.0200). The MediQA verdict breakdown (Figure 3.2) shows that RAG introduces a notable hallucinated segment absent in other conditions.

Table 1: Exact-match accuracy on objective QA datasets. Bold denotes the best result per row.

Dataset	Baseline	CoT	Meta	MedRAG (fixed-k)	MedRAG (adaptive-k)
MedQA-US	0.8174	**0.8783**	0.7609	0.8174	0.8174
PubMedQA	0.2240	0.2520	0.2240	**0.6400**	**0.6400**

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Evaluated using exact-match accuracy.

Table 2: BERTScore F1 on subjective QA datasets. Bold denotes the best result per row.

Dataset	Baseline	CoT	Meta	MedRAG (fixed-k)	MedRAG (adaptive-k)
BioASQ	0.8480	0.7846	0.8227	0.9049	**0.9070**
MediQA	**0.8373**	0.8203	0.8312	0.8344	0.8345

[flushleft]

Evaluated using BERTScore F1.

Table 3: Hallucination rates across datasets and conditions. Bold denotes the lowest (best) rate per row.

Dataset	Baseline	CoT	Meta	MedRAG (fixed-k)	MedRAG (adaptive-k)
BioASQ	0.0882	**0.0147**	0.0882	0.0588	0.0809
MedQA-US	0.1680	**0.0960**	0.1720	0.1720	0.1800
MediQA	0.0240	**0.0080**	0.0200	0.0480	0.0600
PubMedQA	0.4200	0.2040	0.4200	0.1880	**0.1760**

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Lower hallucination rate indicates fewer unsupported responses.

Hallucination is assessed via citation grounding (ROUGE-L recall ≥ 0.15 against retrieved snippets) for MedRAG conditions and against a reference-grounding oracle for non-retrieval conditions.

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Figure 1: Hallucination rate by dataset and condition. CoT consistently achieves the lowest rate on three of four datasets. MedRAG variants reduce hallucination on PubMedQA but increase it on MediQA.

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Figure 2: Raw hallucinated count by dataset and condition ($n = 250$ per dataset, except BioASQ $n = 136$).

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Figure 3: BioASQ verdict breakdown. CoT markedly reduces hallucinated responses but shifts a substantial portion to *unverifiable*, indicating refusals or hedged answers rather than grounded claims.

[width=0.75]dataset$_b reakdown_p ubmedqa.png$

Figure 4: PubMedQA verdict breakdown. MedRAG conditions convert a large share of *hallucinated* responses (Baseline/Meta ≈105) into *supported* ones (≈44–51 hallucinated), while CoT shifts responses toward *unverifiable*.

[width=0.75]dataset$_b reakdown_m ediqa.png$

Figure 5: MediQA verdict breakdown. MedRAG introduces a hallucinated segment (15 responses) not present in Baseline (6) or Meta (5), suggesting that retrieved context can introduce rather than suppress unsupported claims.

[width=0.75]dataset$_b reakdown_m edqa_u s.png$

Figure 6: MedQA-US verdict breakdown. All conditions show broadly similar hallucinated counts; CoT reduces hallucination slightly. The unverifiable segment in CoT reflects responses that cite reasoning chains rather than direct factual snippets.

4 Analysis

4.1 RAG Does Not Uniformly Reduce Hallucination

The headline finding is that retrieval augmentation does not produce consistent hallucination reduction across datasets. MedRAG variants help on PubMedQA (where the task is structurally aligned with retrieved evidence: binary decisions grounded in abstracts) and partially on BioASQ, but they fail on MedQA-US and actively worsen hallucination on MediQA. This heterogeneity suggests that the benefit of RAG is not a property of retrieval per se, but of the match between retrieval format, prompt structure, and question type.

4.2 The MediQA Paradox: When Retrieval Introduces Hallucination

The most striking anomaly is MediQA. MedRAG fixed-k (0.0480) and adaptive-k (0.0600) both double the Baseline hallucination rate (0.0240). The verdict breakdown (Figure 3.2) confirms the pattern: MedRAG introduces 15 hallucinated responses versus 6 for Baseline. Two non-exclusive explanations are plausible.

First, MediQA answers are drawn from a candidate set rather than a single reference string. When retrieved snippets describe phenomena related to but distinct from the expected candidate, the model may incorporate snippet content that is medically plausible but does not match the evaluation oracle, triggering a false hallucination verdict. Second, the retrieved snippets may themselves be noisy or tangentially related to the question, leading the model to blend snippet content with parametric knowledge in ways that reduce rather than improve grounding. Both explanations point to a deeper issue: citation-constrained prompting can make hallucination *worse* when retrieval quality is low or when the evaluation oracle is mismatched with the answer format.

4.3 CoT as Hallucination Suppressor: Mechanism and Limits

CoT achieves the best hallucination rate on BioASQ, MedQA-US, and MediQA. The most likely mechanism is that step-by-step reasoning forces the model to decompose claims before committing to them, creating an implicit self-consistency check. However, the BioASQ and MedQA-US verdict breakdowns (Figures 3.2 and 3.2) reveal a trade-off: CoT increases the *unverifiable* segment. This matters for interpretation. If "unverifiable" corresponds to hedged or abstained responses, then CoT's low hallucination rate partly reflects the model declining to make verifiable claims, not necessarily making fewer false ones. A metric that collapses abstention and grounded support conflates two qualitatively different behaviors.

4.4 Adaptive-k Provides No Consistent Advantage Over Fixed-k

Adaptive-k retrieval was designed to retrieve more snippets for complex queries and fewer for simple ones. In practice, the two MedRAG variants produce nearly identical results across all four datasets—identical on PubMedQA (0.6400 accuracy), within 0.002 BERTScore on BioASQ, and differing by at most 0.008 on hallucination rates. The complexity heuristic (question length, entity count, syntactic structure) may not be an effective proxy for retrieval depth requirements in biomedical QA, where short questions can demand extensive evidence and long questions can be answerable from a single snippet. Absent a systematic evaluation of whether adaptive-k actually varies k in a meaningful way across question types, the claimed benefit cannot be substantiated by the current results.

4.5 Meta Prompting Offers No Reliable Benefit

Meta prompting matches Baseline exactly on BioASQ and PubMedQA hallucination (0.0882 and 0.4200 respectively), degrades accuracy on MedQA-US (0.7609 vs. 0.8174), and provides only marginal hallucination reduction on MediQA (0.0200 vs. 0.0240). The meta prompt used—assigning a biomedical expert persona—appears insufficient to alter the model's grounding behavior. This is consistent with prior work showing that persona-based prompts have limited and inconsistent effects on factual accuracy in capable frontier models.

4.6 Measurement Validity: Citation Grounding as a Hallucination Proxy

A methodological limitation deserves explicit acknowledgment. The hallucination metric is not symmetric across conditions. For MedRAG conditions, "supported" means the generated explanation has ROUGE-L recall ≥ 0.15 against a retrieved snippet. For non-retrieval conditions (Baseline, CoT, Meta), the grounding oracle is different by necessity. This asymmetry means that direct numerical comparisons of hallucination rates across the RAG and non-RAG conditions are not strictly valid: a low hallucination rate for CoT may reflect different evaluative criteria than a low hallucination rate for MedRAG. The results within each condition group (non-RAG vs. RAG) are internally comparable, and cross-condition comparisons should be interpreted with this caveat in mind.

5 Conclusion

This project evaluated five prompting and retrieval conditions—Baseline, CoT, Meta prompting, fixed-k MedRAG, and adaptive-k MedRAG—on four medical QA benchmarks spanning multiple-choice, binary, factoid, and open-response formats. Our results yield three take-aways.

RAG improves accuracy and reduces hallucination selectively, not universally. MedRAG produces large accuracy gains on PubMedQA (+0.39 over Baseline) and strong BERTScore improvements on BioASQ, but provides no accuracy benefit on MedQA-US and actively degrades hallucination on MediQA. The benefit of retrieval augmentation is tightly coupled to whether the task format is structurally compatible with retrieved evidence.

CoT is the most consistent hallucination suppressor. Across three of four datasets, CoT achieves the lowest hallucination rate. Its mechanism appears to be implicit self-consistency from decomposed reasoning. The trade-off is a higher unverifiable rate, which reflects increased hedging rather than improved grounding. Whether hedging is preferable to hallucination in clinical applications is a domain-specific judgment.

Adaptive-k retrieval does not improve over fixed-k in our setting. Despite its theoretical appeal, adaptive-k produces results statistically indistinguishable from fixed-k across all metrics. A more principled complexity estimator—potentially trained on biomedical question difficulty labels—would be needed to realize the expected gains.

Limitations. The hallucination metric is a proxy (ROUGE-L citation grounding) and is not symmetric across retrieval and non-retrieval conditions. BERTScore on MediQA lacks discriminative power due to the candidate-set answer format. The ROUGE-L threshold (0.15) was not empirically calibrated against manual labels, which limits the interpretability of absolute hallucination rates. Future work should employ human annotation or LLM-as-judge evaluation to validate the citation-grounding metric and to provide a common evaluation oracle across all conditions.

References

References follow the acknowledgments in the camera-ready paper. Use unnumbered first-level heading for the references. Any choice of citation style is acceptable as long as you are consistent. It is permissible to reduce the font size to `small` (9 point) when listing the references. Note that the Reference section does not count towards the page limit.

[1] Alexander, J.A. & Mozer, M.C. (1995) Template-based algorithms for connectionist rule extraction. In G. Tesauro, D.S. Touretzky and T.K. Leen (eds.), *Advances in Neural Information Processing Systems 7*, pp. 609–616. Cambridge, MA: MIT Press.

[2] Bower, J.M. & Beeman, D. (1995) *The Book of GENESIS: Exploring Realistic Neural Models with the GEneral NEural SImulation System.* New York: TELOS/Springer–Verlag.

[3] Hasselmo, M.E., Schnell, E. & Barkai, E. (1995) Dynamics of learning and recall at excitatory recurrent synapses and cholinergic modulation in rat hippocampal region CA3. *Journal of Neuroscience* **15**(7):5249-5262.

A Technical appendices and supplementary material

Technical appendices with additional results, figures, graphs, and proofs may be submitted with the paper submission before the full submission deadline (see above). You can upload a ZIP file for videos or code, but do not upload a separate PDF file for the appendix. There is no page limit for the technical appendices.

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

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